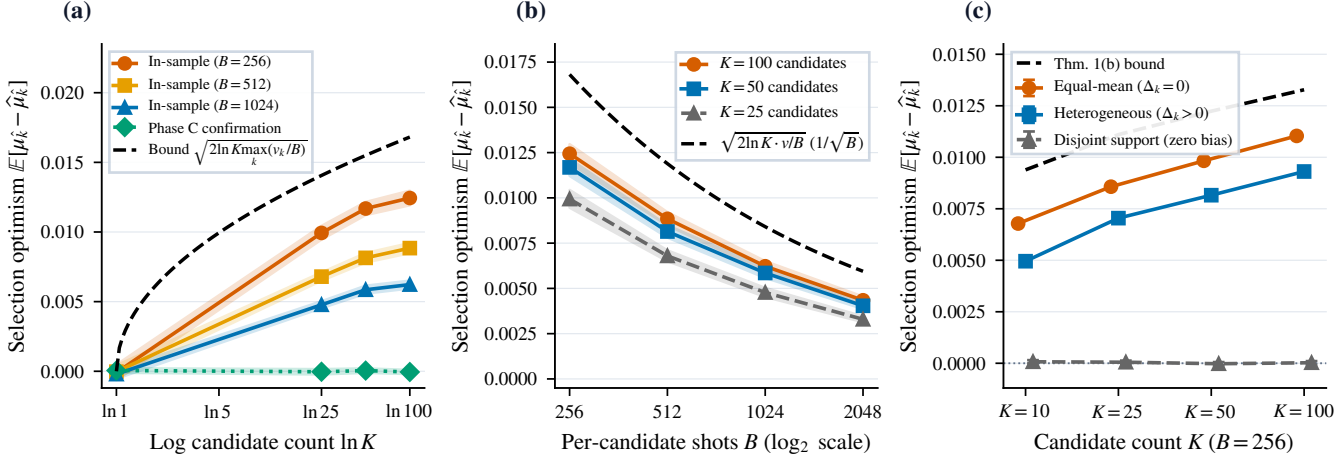
(a)
In-sample ($B = 256$)
In-sample ($B = 512$)
In-sample ($B = 1024$)
Phase C confirmation
Bound $\sqrt{2 \ln K \max_k (v_k / B)}$
Selection optimism $\mathbb{E}[\mu_{\hat{k}} - \hat{\mu}_{\hat{k}}]$
0.020
0.015
0.010
0.005
0.000
ln 1
ln 5
ln 25
ln 100
Log candidate count ln K
(b)
$K = 100$ candidates
$K = 50$ candidates
$K = 25$ candidates
$\sqrt{2 \ln K \cdot v / B}$ ($1/\sqrt{B}$)
Selection optimism $\mathbb{E}[\mu_{\hat{k}} - \hat{\mu}_{\hat{k}}]$
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
256
512
1024
2048
Per-candidate shots B ($\log_2$ scale)
(c)
Thm. 1(b) bound
Equal-mean ($\Delta_k = 0$)
Heterogeneous ($\Delta_k > 0$)
Disjoint support (zero bias)
Selection optimism $\mathbb{E}[\mu_{\hat{k}} - \hat{\mu}_{\hat{k}}]$
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
$K = 10$
$K = 25$
$K = 50$
$K = 100$
Candidate count K ($B = 256$)